

7.1.3 Software Architectural Design Process

7.1.3.1 Purpose

The purpose of the Software Architectural Design Process is to provide a software design that implements the software requirements and can be verified and validated against those requirements.

7.1.3.2 Outcomes

As a result of successful implementation of the Software Architectural Design Process:

- a) A software architectural design is developed that describes the software items implementing the software requirements.
 - b) Internal and external interfaces of each software item are defined.
 - c) Consistency and traceability are established between software requirements and software design.
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7.1.3.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures.

7.1.3.3.1 Software Architectural Design Development

This activity develops and documents the architectural structure of the software, including interfaces and databases.

- **7.1.3.3.1.1** Develop and document the software architectural design for each software item or configuration item, if identified.
- **7.1.3.3.1.2** Develop and document a top-level design for interfaces external to the software item.
- **7.1.3.3.1.3** Develop and document a top-level design for the interfaces between software components of the software item.
- **7.1.3.3.1.4** Develop and document a top-level design for the database associated with the software item.

- **7.1.3.3.1.5** Develop and document preliminary user documentation.
 - **7.1.3.3.1.6** Define and document preliminary test requirements and schedules for software integration activities.
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7.1.3.3.2 Evaluation and Review of Architectural Design

This activity evaluates the architectural design to ensure compliance, consistency, and feasibility.

- **7.1.3.3.2.1** Evaluate the software item's architecture, interface, and database designs against the following criteria:
 - a) Traceability to the software item's requirements.
 - b) External consistency with the software item's requirements.
 - c) Internal consistency between software components.
 - d) Appropriateness of design methods and standards.
 - e) Feasibility of detailed design.
 - f) Feasibility of operation and maintenance.
 - **7.1.3.3.2.2** Conduct formal review(s) of the software architectural design in accordance with the **Software Review Process** (7.2.6).
 - **7.1.3.3.2.3** Record and control outputs and baselines through the **Software Documentation Management Process** (7.2.1) and the **Software Configuration Management Process** (7.2.2).
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Cross-References

This process interacts with the following:

- **Software Requirements Analysis Process (7.1.2)** — for traceability and input requirements.
- **Software Detailed Design Process (7.1.4)** — for elaboration of the architecture into unit-level design.
- **Software Documentation Management Process (7.2.1)** — for documenting architectural artifacts.
- **Software Review Process (7.2.6)** — for formal reviews.